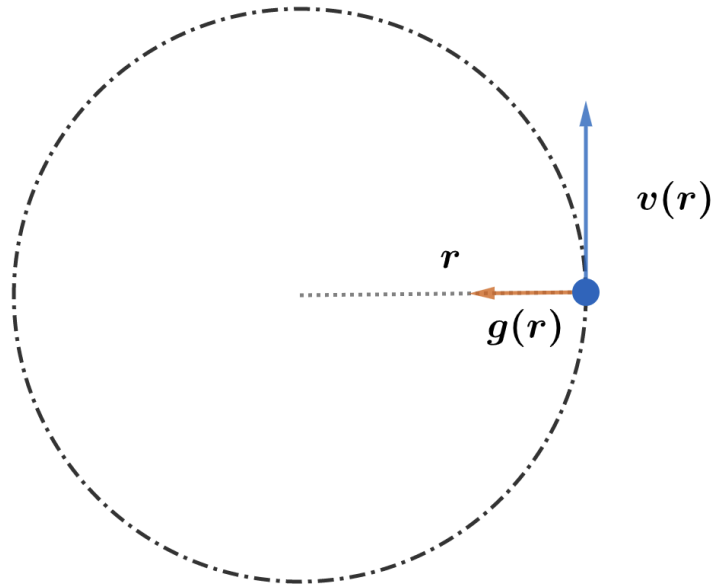


2023 F=ma Exam: Problem 15

Kevin S. Huang



By Gauss's law for gravity, we have

$$\Phi_g = -4\pi GM_{\text{enc}}$$

The flux $\Phi_g \propto g(r) \cdot r^2$ where $g(r)$ is the gravitational field at radius r so

$$g(r)r^2 \propto M(r)$$

$$g(r) \propto \frac{M(r)}{r^2}$$

where $M(r)$ is the mass enclosed in a sphere of radius r . The gravitational force provides the centripetal acceleration so we have

$$mg(r) = \frac{mv(r)^2}{r}$$

$$v(r)^2 = rg(r) \propto \frac{M(r)}{r}$$

Thus,

$$M(r) \propto rv(r)^2$$

Since $v(r) \sim \text{const}$ from the plot, $M(r) \propto r$ so mass enclosed increases linearly with the radius. The answer is B.